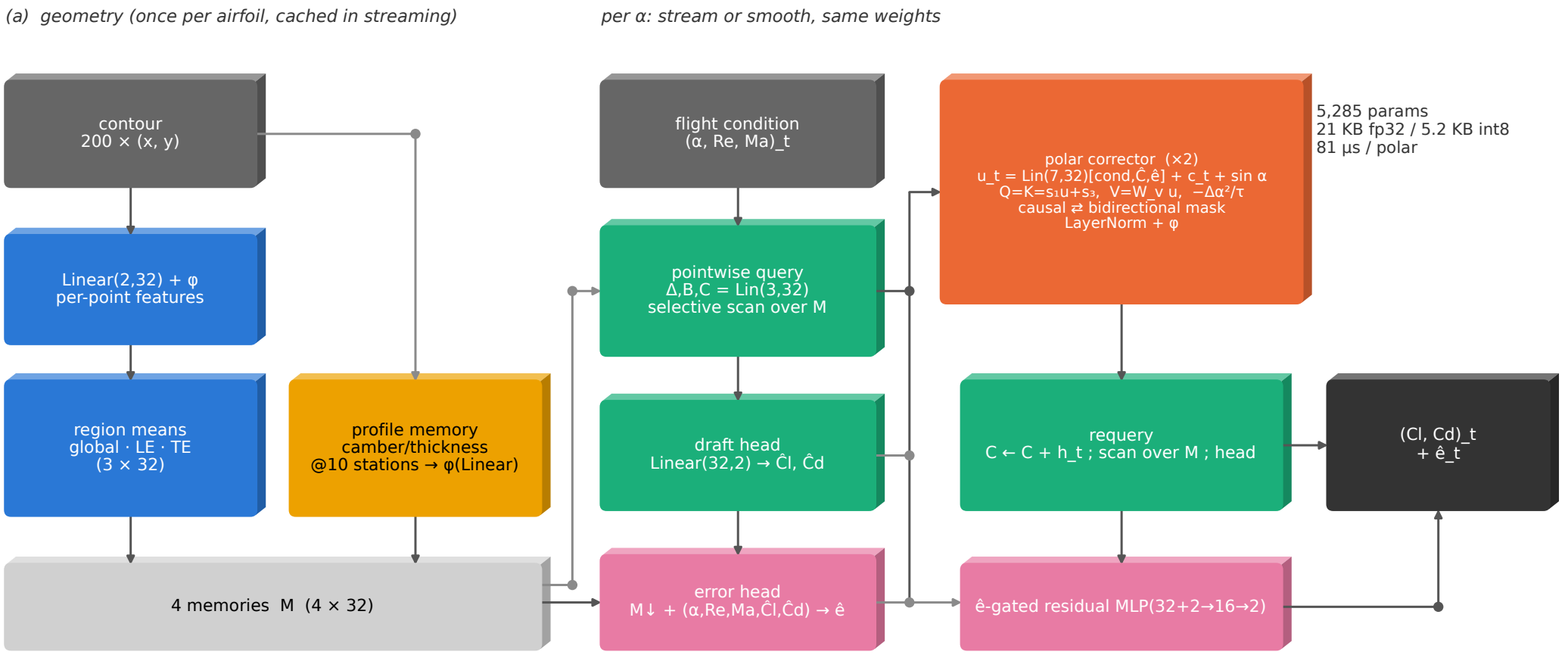
(a) geometry (once per airfoil, cached in streaming)
per α: stream or smooth, same weights
contour
200 × (x, y)
Linear(2,32) + φ
per-point features
region means
global · LE · TE
(3 × 32)
profile memory
camber/thickness
@10 stations → φ(Linear)
4 memories M (4 × 32)
flight condition
(α, Re, Ma)_t
pointwise query
Δ,B,C = Lin(3,32)
selective scan over M
draft head
Linear(32,2) → Ĉl, Ĉd
error head
M↓ + (α,Re,Ma,Ĉl,Ĉd) → ê
polar corrector (×2)
u_t = Lin(7,32)[cond,Ĉ,ê] + c_t + sin α
Q=K=s₁u+s₃, V=W_v u, −Δα²/τ
causal ⇄ bidirectional mask
LayerNorm + φ
requery
C ← C + h_t ; scan over M ; head
ê-gated residual MLP(32+2→16→2)
(Cl, Cd)_t
+ ê_t
5,285 params
21 KB fp32 / 5.2 KB int8
81 µs / polar